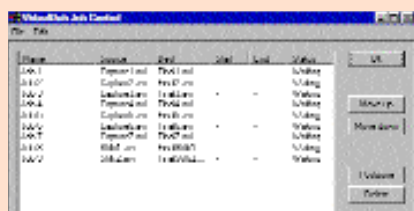


Putting it all Together

You have video clips, sound and stills ready for the final movie. It's time to bring it all together.

Batch convert: If you have several raw video clips and those made from still images, you need to join them to make the final video. To be able to do this, you will have to ensure that they are all at the same resolution, frame rate and compression. If you need to convert the videos to a single format, you can set up a batch conversion process in VirtualDub.

Set up the video settings and select **File > Save as AVI**. Select the box next to 'Add operation to job list and defer processing'.

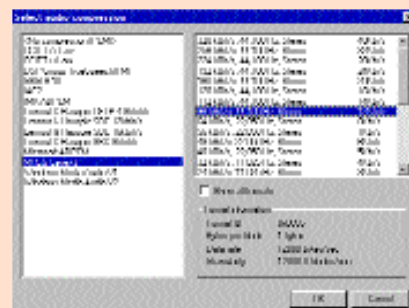


Set up batch processing in VirtualDub

Repeat the same for every file you want to convert. VirtualDub will create a batch file for every job added this way. Open **File > Job control** to bring up the list of jobs. You can delete or postpone jobs from here, or simply click Start to begin converting.

Join video clips: This is the simplest process. Open the first video file in the sequence, then click **File > Append AVI segment**. Select the next file and repeat—you can run the entire video as a single file. You can edit it and apply filters the same way as you would to individual files. If you are not editing it, select **Video > Direct stream copy**. VirtualDub will simply copy all the frames from the video clips to the output file, and not process each one. This is extremely fast. You can save the video here or jump to the next step of merging the sound.

Integrating sound: Once you have the wave file of the same length as the movie, in VirtualDub click **Audio > WAV Audio** to select the sound source for the video. Click **Audio > Full processing mode** and open



Compress audio when integrating with the video to reduce file size

Audio > Compression. You can select any audio codec to compress the sound, but MP3 is the best as it gives the best quality sound and the highest compression.

You can also adjust the volume level of the audio from **Audio > Volume**, though it would be much better to do this in a sound-editing application like SoundForge. When you save the AVI, it will have the sound integrated into it.

One of the first video capturing hardware to reach the masses was the TV Tuner card with capture functionality. On many TV Tuner cards, the capture is software driven. So, even if the software bundled with the card does not allow you to capture the video, other tools should be able to do so. Then there are capture cards, which interface to various video input sources. For example, these take TV or VCR signals as input, which means you can record your favourite shows on television or convert your old tapes to digital format. TV Tuner and capture cards can only convert an incoming video stream and cannot be used to shoot a live video.

One of the best options for amateurs is the Web cam—they are very affordable and have multiple applications such as video and still capture, videoconferencing and some can even be set up as surveillance systems. Web cams do have some drawbacks, though. They generally do not grab good quality images. Also, the entire processing of the video happens on the computer, not the camera, hence, if the system is slow, the video quality is further affected. Again, a Web cam is useful only for live video capture.

A powerful machine with a fast processor, loads of RAM and gigabytes of hard disk space will yield a much smoother movie authoring process, but even a Pen-

tium 200 system with 128 MB of RAM will do. What you compromise on the power of the system, you have to make up for with patience.

For software, you can use the proprietary software bundled with the hardware, or freeware such as VirtualDub. Pinnacle Studio 7 (www.pinnaclesys.com) provides a complete capturing and editing suite. It is especially useful for professional effects and is yet simple enough for anyone to use. It can take input from several sources and output it to a CD or even DVD. Similarly, for audio capture and editing you can use software bundled with the soundcard, or tools like SoundForge (www.sonicfoundry.com) or Cool Edit Pro (www.syntrillium.com). These are all very easy to use, unlike some of the more professional software.

As a rule of thumb, unless you're making video on the spot, have a script ready before you begin shooting. Knowing how you want the video to progress helps tremendously and you won't have any missing elements when you are editing the video clips.

Looking ahead

With the power of video on your desktop, you can wave your magic mouse and just let your imagination run wild. Convert your old marriage tapes into digital video,

add captions to all the funny moments, burn them to VCDs or even DVDs, or impress your friends with your own version of *The Matrix*.

With broadband slowly encroaching upon us, soon you will be hosting your videos on the Web. The possibilities are endless. You may also be able to share a 'live telecast' of your son's birthday bash in India with his grandmother in the US. Or stream a conference from your head office in Delhi to regional offices across the world. TV channels should also start telecasting shows on the Internet. Initially they would only telecast select items like cricket matches, but one day they may choose to run the entire channel online.

The digital video scene is only going to get better. Expect far more powerful, cheaper products and much easier tools in the near future. Some motherboards already integrate capture capabilities onboard. Processors themselves are evolving at break-neck speed. Even with the products available today, you can capture at DVD resolutions. Encoding Dolby Digital sound on the desktop was not possible until very recently—it is now.

The tools of today allow professional quality multimedia authoring. The tools of tomorrow promise to be even better. This is not magic. This is reality. 

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